

2022 midterm geometry

Part I.

$K = (0, i, j)$, $K' = (0', i', j')$; $K \rightarrow$ right-oriented orthonormal sys.

$$[0']_K = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$$[i']_K = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$[j']_K = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\Delta ABC: [A]_K = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, [B]_K = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, [C]_K = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

P1) $\vec{BC}$, $i \rightarrow$ orthogonal to each other?

$$\vec{BC} = C - B = \begin{bmatrix} 4 \\ 2 \end{bmatrix} - \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$K \text{ orthonormal system} \Rightarrow i = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{BC}, i \rightarrow \text{orthogonal} \Leftrightarrow \vec{BC} \cdot i = 0 \Leftrightarrow \langle \vec{BC}, i \rangle = 0$$

$$\Leftrightarrow (0, 1) \quad \Leftrightarrow (0 \cdot 1) + (1 \cdot 0) = 0$$

$$\Leftrightarrow 0 = 0 \quad \checkmark$$

P2) $d(B, AC) = ?$ CARTESIAN EQ. OF AC

$$d(P, H) = \frac{|ap_1 + bp_2 - c|}{\sqrt{a^2 + b^2}}$$

$P(p_1, p_2)$

$$\text{line: } ax + by + c = 0$$

$$\vec{AC} = C - A = \begin{bmatrix} 2 \\ -2 \end{bmatrix} \text{ direction vector}$$

$$\Rightarrow m = (1, 1) \text{ normal vector}$$

$$\hookrightarrow (1, 1) \cdot (2, -2) = 2 - 2 = 0$$

$$\Rightarrow AC: x + y + c = 0$$

$$A \in AC \Rightarrow 2 + 4 + c = 0 \Rightarrow c = -6$$

$$A(2, 4)$$

$$\Rightarrow \boxed{AC: x + y - 6 = 0} \text{ eq. of the line.}$$

$$\begin{aligned} \Rightarrow a &= 1 \\ \Rightarrow b &= 1 \\ \Rightarrow c &= -6 \end{aligned}$$

$$d(B, AC) = \frac{|1 \cdot 4 + 1 \cdot 1 - 6|}{\sqrt{1^2 + 1^2}} = \frac{|-1|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$B(4, 1)$$

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P3) $H \triangle ABC \rightarrow$ orthocenter? $\rightarrow K$.

$$\vec{AC} = \begin{bmatrix} 2 \\ -2 \end{bmatrix} \Rightarrow AC \text{ has direction } (1, -1).$$

$d(B, AC)$:

$\perp$ direction: $(1, 1)$

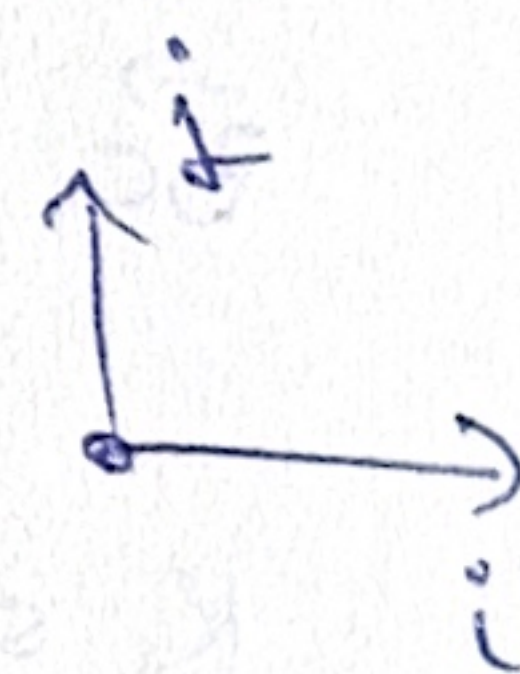
$$(\text{because } (1, -1) \cdot (1, 1) = 1 \cdot 1 + (-1) \cdot 1 = 1 - 1 = 0 \Rightarrow \perp).$$

$$l_B: (x, y) = \underset{\substack{\downarrow \\ B}}{(4, 1)} + t \cdot \underset{\substack{\downarrow \\ \text{direction} \\ \text{vector}}}{(1, 1)}$$

$$\Rightarrow \begin{cases} x = 4 + t \Rightarrow t = x - 4 \\ y = 1 + t \Rightarrow y = 1 + (x - 4) = x - 3 \end{cases} \Rightarrow \boxed{y = x - 3}$$

$d(A, BC)$:

$$\vec{BC} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow BC \text{ vertical } (j \neq 0, i = 0)$$



$$l_A: (x, y) = (2, 4) + t \cdot (1, 0)$$

$$\Rightarrow \begin{cases} x = 2 + t \\ y = 4 \end{cases} \Rightarrow \boxed{y = 4}$$

$$n: \begin{cases} y = x - 3 \\ y = 4 \end{cases} \Rightarrow x - 3 = 4 \Rightarrow x = 7, y = 4 \Rightarrow H(7, 4) \\ \Rightarrow [H]_K = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$$

P4) AC w.r. to K' .
(eq.)

$$\boxed{[P]_{K'} = M_{B'B}([P]_K - [O']_K)}$$

$$M_{BB'} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \quad \begin{matrix} \downarrow & \downarrow \\ i' & j' \end{matrix} \quad \leadsto M_{B'B} = M_{BB'}^{-1} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\text{Aug } \left[\begin{array}{cc|cc} -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_1 + R_2 \rightarrow R_2} \left[\begin{array}{cc|cc} -1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{array} \right] \xrightarrow{R_1 \cdot (-1) \rightarrow R_1} \left[\begin{array}{cc|cc} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 1 \end{array} \right]$$

$$\begin{bmatrix} -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \cdot (-1) \rightarrow R_1} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$\underbrace{\quad}_{I_2} \quad \underbrace{\quad}_{\text{inv.}}$

2f

$$\begin{aligned} [A]_{K'} &= M_{B'B}^{-1} ([A]_K - [O']_K) \\ &= \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \cdot \left(\begin{bmatrix} 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 4 \\ 0 \end{bmatrix} \right) \\ &= \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 2 \end{bmatrix} \end{aligned}$$

$$\Rightarrow A_{K'} = (2, 2).$$

$$\begin{aligned} [C]_{K'} &= M_{B'B}^{-1} ([C]_K - [O']_K) \\ &= \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \cdot \left(\begin{bmatrix} 4 \\ 2 \end{bmatrix} - \begin{bmatrix} 4 \\ 0 \end{bmatrix} \right) \\ &= \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 2 \end{bmatrix} \end{aligned}$$

$$\Rightarrow C_{K'} = (0, 2)$$

$$A_C : ax + by + c = 0$$

How to find line $A_{C'}$

$$\begin{cases} A'(2, 2), C'(0, 2) \\ \vec{A'C'} = C' - A' = (0, 2) - (2, 2) = (-2, 0) \text{ dir. vector} \\ (x', y') = (2, 2) + t \cdot (-2, 0) \Rightarrow \begin{cases} x' = 2 - t \\ y' = 2 \end{cases} \Rightarrow \boxed{y' = 2} \end{cases}$$

$\downarrow \quad \quad \downarrow$
 $A \quad \quad \text{dir.}$

Cartesian form:
 $\boxed{y' - 2 = 0}$

Part II:

$K = (O, i, j, k)$ right-oriented orthonormal sys.

ABCD tetrahedron.

$$[A]_K = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, [B]_K = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, [C]_K = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}, [D]_K = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

K' : coord. system $(D, \vec{BA}, \vec{DA}, \vec{DC})$.

P5) $d(A, BD)$

$$\vec{BD} = D - B = (1, 1, 1) - (1, 2, 1) = (0, -1, 0).$$

$$(x, y, z) = (1, 2, 1) + t \cdot (0, -1, 0)$$

$$\Rightarrow \begin{cases} x = 1 \\ y = 2 - t \\ z = 1 \end{cases} \Rightarrow \text{every point on } BD \text{ has: } \begin{cases} x = 1 \\ z = 1 \end{cases}$$

$$A(1, 2, 0)$$

$$B(1, 2, 1)$$

$$\vec{AB} = (1, 2, 1) - (1, 2, 0) = (0, 0, 1)$$

$$|\vec{AB}| = \sqrt{0^2 + 0^2 + 1^2} = 1.$$

$$\Rightarrow d(A, BD) = 1.$$

P6) w.r. to K , orthogonal proj. of A on the plane containing $\triangle BCD$

$$B(1, 2, 1)$$

$$(1, 2, 1) \cdot (1, 0, 1) = (1 \cdot 1) + 2 \cdot 0 + 1 \cdot 1$$

$$\vec{BC} = C - B = (-1, 1, 1)$$

$$\vec{BD} = D - B = (0, -1, 0).$$

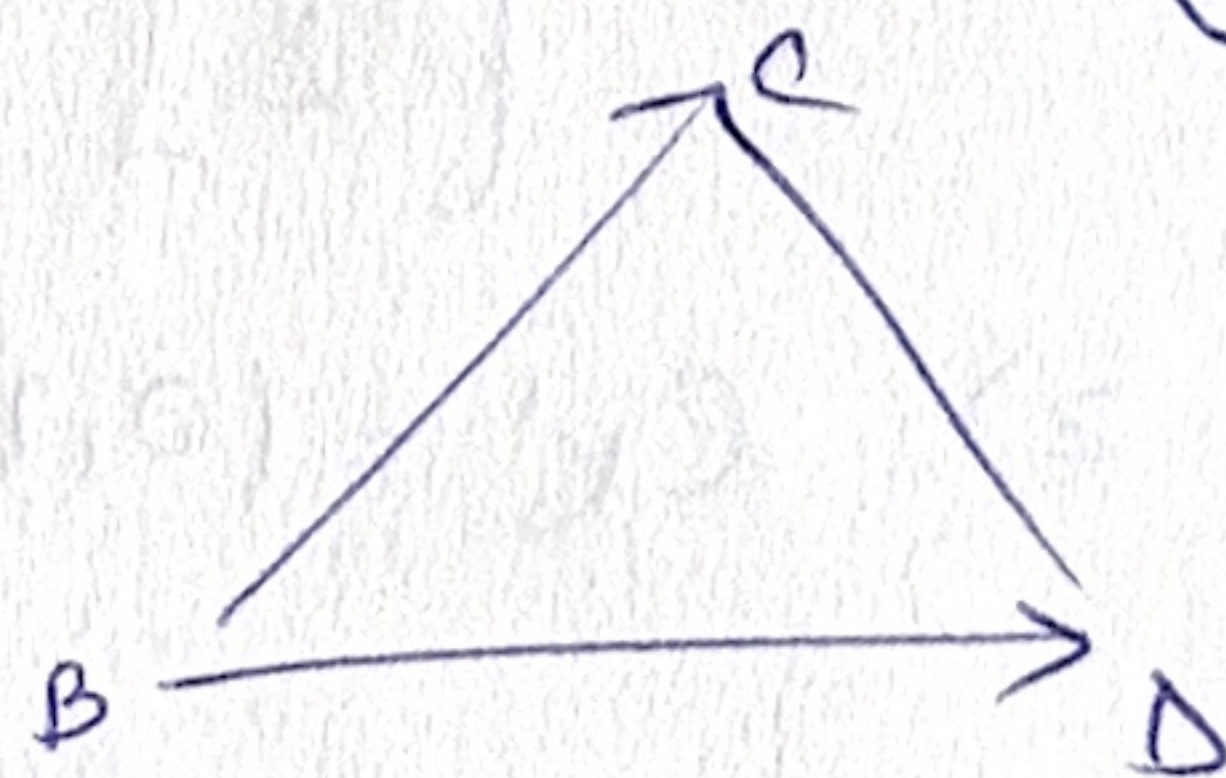
$$\vec{n} = \vec{BC} \times \vec{BD} \text{ normal vector to the plane}$$

$$(-1, 1, 1) \times (0, -1, 0) = (1 \cdot 0) + (1 \cdot (-1)) + 1 \cdot 0 = 0 - 1 = -1.$$

$$= \begin{vmatrix} i & j & k \\ -1 & 1 & 1 \\ 0 & -1 & 0 \end{vmatrix} = i \cdot \begin{vmatrix} 1 & 1 \\ -1 & 0 \end{vmatrix} - j \cdot \begin{vmatrix} -1 & 1 \\ 0 & 0 \end{vmatrix} + k \cdot \begin{vmatrix} -1 & 1 \\ 0 & -1 \end{vmatrix} =$$

$$= i(0 + 1) - j(0) + k(1 - 0) = i + k = (1, 0, 1).$$

$$\Rightarrow \vec{n} = (1, 0, 1).$$



$$3(1,2,1), \vec{m} = (1,0,1).$$

$$1 \cdot (x-x_B) + 0 \cdot (y-y_B) + 1 \cdot (z-z_B) = 0.$$

$$\Rightarrow (x-1) + 0(y-2) + (z-1) = 0$$

$$\Rightarrow x-1+z-1=0$$

$$\Rightarrow x+z-2=0.$$

$$\Rightarrow \underline{x+z=2.} \quad \text{eq. of plane BCD.}$$

H: projection point, on the line through A $\perp \vec{m}$.

$$H = A + t \cdot (1,0,1)$$

$$H = (1,2,0) + t \cdot (1,0,1)$$

$$H = \begin{pmatrix} 1+t \\ 2 \\ t \end{pmatrix} \in \text{plane BCD.}$$

$$\Rightarrow (1+t) + t = 2$$

$$\Rightarrow 1+2t=2$$

$$\Rightarrow 2t=1 \Rightarrow t=\frac{1}{2}$$

$$H = \left(1+\frac{1}{2}, 2, \frac{1}{2}\right) = \left(\frac{3}{2}, 2, \frac{1}{2}\right)$$

P4) w.k. to K, $(\vec{AB} \times \vec{AC}) \times K$.

$$\vec{AB} = B-A = (1,2,1) - (1,2,0) = (0,0,1).$$

$$\vec{AC} = C-A = (0,3,2) - (1,2,0) = (-1,1,2).$$

$$\begin{aligned} \vec{AB} \times \vec{AC} &= (0,0,1) \times (-1,1,2) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ -1 & 1 & 2 \end{vmatrix} = 1 \cdot (-1)^5 \cdot \begin{vmatrix} \hat{i} & \hat{j} \\ -1 & 1 \end{vmatrix} \\ &= -(i+j) = -i-j = (-1,-1,0). \end{aligned}$$

$$\begin{aligned} (\vec{AB} \times \vec{AC}) \times K &= (-1,-1,0) \times (0,0,1) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \\ &= 1 \cdot (-1)^6 \cdot \begin{vmatrix} \hat{i} & \hat{j} \\ -1 & -1 \end{vmatrix} = -i+j = (-1,1,0). \end{aligned}$$

P8) K' - left / right oriented?

$$K': (D, \vec{BA}, \vec{DA}, \vec{DC}).$$

$$\vec{BA} = A - B = (1, 2, 0) - (1, 2, 1) = (0, 0, -1).$$

$$\vec{DA} = A - D = (1, 2, 0) - (1, 1, 1) = (0, 1, -1).$$

$$\vec{DC} = C - D = (0, 3, 2) - (1, 1, 1) = (-1, 2, 1).$$

$$\Rightarrow M = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 2 \\ -1 & -1 & 1 \end{bmatrix}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $\vec{BA} \quad \vec{DA} \quad \vec{DC}$

$$\det M = \begin{vmatrix} 0 & 0 & -1 \\ 0 & 1 & 2 \\ -1 & -1 & 1 \end{vmatrix} = (-1) \cdot (-1)^4 \cdot \begin{vmatrix} 0 & 1 \\ -1 & -1 \end{vmatrix} = - (0 + 1) = -1$$

$\Rightarrow K'$ has opposite orientation to K which is right-orient (basis)

$\Rightarrow K'$ is left-orient.

P9) w.r. to K' , eq. of plane which contains $\triangle ABC$.

$$K': (D, \vec{BA}, \vec{DA}, \vec{DC}).$$

$$\vec{DC} = x' \cdot \vec{BA} + y' \cdot \vec{DA} + z' \cdot \vec{DC}$$

$$K': (x', y', z').$$

$$\Rightarrow [C]_{K'} = (0, 0, 1).$$

$$\vec{DA} = x' \cdot \vec{BA} + y' \cdot \vec{DA} + z' \cdot \vec{DC}$$

$$\vec{DA} = x' \cdot \vec{BA} + y' \cdot \vec{DA} + z' \cdot \vec{DC} \Rightarrow y' = 1. \Rightarrow [A]_{K'} = (0, 1, 0).$$

$$\vec{DB} = x' \cdot \vec{BA} + y' \cdot \vec{DA} + z' \cdot \vec{DC} = (0, 1, 0)$$

$$\vec{DB} = B - D = (1, 2, 1) - (1, 1, 1) = (0, 1, 0)$$

$$(0, 1, 0) = x' \cdot (0, 0, -1) + y' \cdot (0, 1, -1) + z' \cdot (-1, 2, 1)$$

$$\Rightarrow (0, 1, 0) = (-z', y' + 2z', -x' - y' + z')$$

$$\Rightarrow \begin{cases} -z' = 0 \\ y' + 2z' = 1 \Rightarrow y' = 1 \\ -x' - y' + z' = 0 \Rightarrow x' + y' = 0 \Rightarrow x' = -1 \end{cases}$$

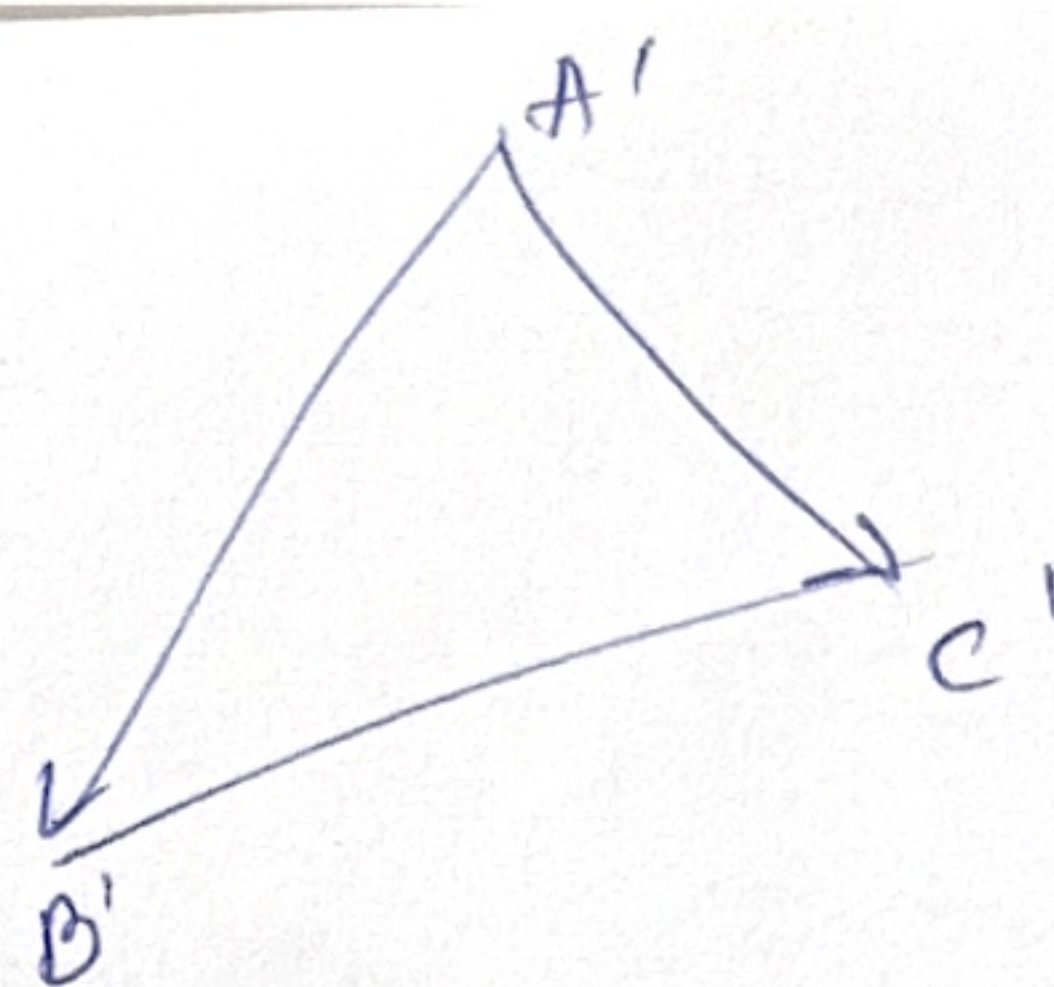
$$\Rightarrow (-1, 1, 0) \Rightarrow [B]_{K'} = (-1, 1, 0). \quad \textcircled{6}$$

$$A'(0,1,0), B'(-1,1,0), C'(0,0,1).$$

$$\vec{A'B'} = (-1, 0, 0)$$

$$\vec{A'C'} = (0, -1, 1)$$

$$\vec{m'} = \vec{A'B'} \times \vec{A'C'} = (-1, 0, 0) \times (0, -1, 1)$$



$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 0 & 0 \\ 0 & -1 & 1 \end{vmatrix} = (-1)^3 \begin{vmatrix} \vec{j} & \vec{k} \\ -1 & 1 \end{vmatrix} = +1(\vec{j} + \vec{k}) = +\vec{j} + \vec{k} = (0, +1, +1).$$

$\vec{BA},$
 $\vec{CA},$
 $\vec{BC}$

$$\Rightarrow \text{plane eq: } 0(x' - x_{A'}) + 1 \cdot (y' - y_{A'}) + 1 \cdot (z' - z_{A'}) = 0$$

$$\Rightarrow \cancel{0}(x' - 0) + 1 \cdot (y' - 1) + 1 \cdot (z' - 0) = 0$$

$$\Rightarrow y' - 1 + z' = 0$$

$$\Rightarrow y' + z' - 1 = 0.$$

$$\Rightarrow \underline{y' + z' = 1}.$$